# UPKAR KUMAR

## GenAI & Distributed Systems

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## PROFESSIONAL SUMMARY

- With 12+ years of proven success architecting and scaling distributed systems, high-scale microservices, and GenAI platforms on cloud infrastructure — prioritizing performance, scalability, and robust system design.
- Recognized expertise in implementing cutting-edge LLM technologies — RAG frameworks, Agentic AI, LLM Gateways, and MLOps pipelines — to enhance software solutions at scale across multi-product SaaS platforms.
- Demonstrated proficiency in stakeholder management and cross-functional technical leadership — driving org-wide architecture standards, authoring RFCs and ADRs, and ensuring engineering alignment across 10+ teams.
- Adept at applying software design principles, event-driven patterns, and agile methodologies to deliver state-of-the-art, extensible system designs that translate ambiguous product requirements into production-ready platforms.
- Committed to engineering excellence — observable systems, rigorous load testing, CI/CD automation, and data-driven performance optimization.

## PROFESSIONAL EXPERIENCE

### Tech Lead

**Freshworks** • *Mar 2022 – Present,* Bangalore, India

**Architecting the Agentic Platform & Native Assistant Framework**

- Led the architectural vision and end-to-end roadmap for the Agentic Platform — designing the full system from intent to execution, covering agent orchestration, tool invocation, knowledge base retrieval, and multi-turn conversation management across Freshworks products.
- Architected the strategic migration from Azure OpenAI Assistants API to a proprietary in-house framework — eliminating vendor lock-in, cutting per-session inference cost by ~55%, and maintaining full API contract compatibility so zero upstream changes were required; adopted across 4 product lines within 3 months.
- Designed a composable agent execution engine using Flowise and LangGraph: ReAct-style reasoning loops, tool invocation, streaming responses, stateful thread management — each capability an independently deployable building block.
- Established the GenAI Platform Service Contract and extensible plugin architecture — adopted by 8+ feature teams as the unified, versioned API, eliminating bespoke LLM integrations across Freshworks. Implemented full observability per agent run: every reasoning step, tool call, latency, and outcome captured.
- Implemented safety and governance layers: input/output validation, PII scrubbing, per-tenant rate limiting, graceful fallback handling — ensuring reliable, secure agent behaviour in customer-facing environments.

◆ **Tech:** Java 21, Spring Boot, Flowise, LangGraph, Azure OpenAI, OpenSearch, Redis, Kafka, Prometheus, Grafana, Kubernetes, Istio

**Architecting AI-Driven Insights & Knowledge Gap Detection**

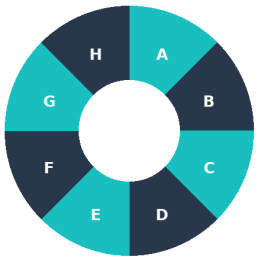
- Architected the Insights & Knowledge Gap platform, delivering AI-generated natural language business insights — trend analysis, anomaly detection, root cause summaries — from product telemetry data across all Freshworks products, serving 5000+ business users.
- Designed the Knowledge Gap Detection engine: leverages human-agent resolutions as ground truth to automatically identify missing, conflicting, or outdated KB content — surfaced actionable gaps in 30%+ of existing articles, reducing content maintenance effort by ~50%.
- Consolidated the MLOps data pipeline: AWS DMS (CDC) → Apache Iceberg (S3) → AWS Glue orchestration → Athena for near-real-time feature engineering with strict multi-tenant data isolation.
- Established LLM-as-judge + human feedback evaluation pipelines to continuously measure insight quality, factual accuracy, and relevance — closing the feedback loop into prompt tuning and model selection cycles.

◆ **Tech:** Java 21, Python, Spring Boot, Kafka, Redis, OpenSearch, Azure OpenAI, AWS Glue, Apache Iceberg, AWS Athena, AWS DMS, Prometheus, Grafana, Kubernetes

**Architecting Freddy AI Platform (LLM Gateway with Dialogue Manager)**

- Designed and deployed Freshworks' company-wide LLM Gateway processing 5M+ calls/day — a unified product-agnostic API across Azure OpenAI (GPT-5/4), AWS Bedrock

## MY TIME




**A** Coding
**B** Code Review
**C** System Design
**D** Design Review
**E** Mentoring Peers
**F** Pair Programming
**G** Stakeholder Interaction
**H** Perf & Scalability

## SKILLS

**Languages & Frameworks**
**Java 21 Python JavaScript Spring Boot FastAPI React**
**GenAI & ML**
**LLM / RAG / Agents Azure OpenAI AWS Bedrock Flowise LangGraph**
**Data & Messaging**
**Kafka OpenSearch Redis Iceberg Glue Athena DMS MySQL MongoDB**
**Infra & Cloud**
**K8s Docker Istio AWS Azure Jenkins CI/CD OpenTelemetry**
**Architecture**
**Microservices Event-Driven System Design**
**Observability**
**Prometheus Grafana**

## AWARDS & RECOGNITION

★ **Aegis Graham Bell – Finalist**
Innovation in Enterprise Integration (SAP)

★ **Best Team Player**
Freshworks

★ **COVID-19 Open Source**
karnatakafightscorona.org

## PERSONAL PROJECTS

**Stock Trader – Options Bot**
Aggregates real-time news, financials & macro signals to autonomously trade options — auto-sets stop-loss & GTT. Multi-user with per-user watchlist and risk config.

**Vidya – Talk to Video**
AI tutor Chrome extension: ask doubts by voice while watching any video; answers using only that video's context, returns timestamp to jump to.

**Best Route Suggester**
Finds confirmed train tickets across multi-hop, break journey & nearby

(Claude, Titan), and internal models. Intelligent cost/latency-based routing reduced total AI spend by ~40%; onboarded 8 product teams in the first quarter.

- Formalized the 3-year LLM Platform technical roadmap, authoring RFCs and ADRs to institutionalize key design decisions and ensure long-term architectural alignment for 10+ teams across Freshworks.
- Designed Freshworks' first-generation Dialogue Manager prior to the industry-wide availability of native LLM tool-calling — a custom Intent Handler Registry Protocol enabling intent-based dispatch, serving as an internal precursor to MCP. The dialogue manager maintained multi-turn conversation state, tracked context across turns, and powered Freshworks Copilot across all products.
- Built robust governance: semantic response caching (Redis), prompt versioning, A/B testing, and shadow mode for safe model rollouts; employed OpenTelemetry for custom metrics, enhancing observability through Prometheus and Grafana. Rigorous load testing optimized high throughput and low latency using Java 21 virtual threads.

◆ **Tech:** Java 21, Spring Boot, Azure OpenAI, AWS Bedrock, Redis, Kafka, Prometheus, Grafana, OpenTelemetry, Kubernetes, Istio

**Architecting MLOps Middleware – Auto Triage & Canned Response**

- Architected a real-time ML middleware achieving 87% auto-triage accuracy in production — classifying support tickets by category, priority, and product area, reducing manual triage overhead by ~65% and cutting average first-response time by ~35%.
- Engineered the canned response generation pipeline: leverages historical resolution patterns and LLMs to suggest contextually accurate, tone-appropriate agent responses — improving efficiency at scale.
- Designed the end-to-end MLOps data pipeline: AWS DMS (CDC) → Apache Iceberg (S3) via AWS Glue → Athena for feature engineering and data science experimentation without impacting production. Built data drift detection pipelines triggering automated Airflow retraining workflows when classification accuracy degrades below threshold.

◆ **Tech:** Java 21, Python, Kafka, Apache Iceberg, AWS Glue, AWS Athena, AWS DMS, MLFlow, Airflow, Prometheus, Grafana, Kubernetes

**Architecting the Unified Conversation Service**

- Architected Freshworks' unified Conversation Service — the foundational platform managing all conversational transactions across Freshchat, Freshdesk, and Freshservice at 50K+ events/sec peak write throughput, serving 10,000+ business customers.
- Designed the event-sourced ingestion pipeline (Kafka → DocumentDB) with a read-optimized OpenSearch layer; architected conversation threading, participant schema, and full audit trail — maintaining p99 read latency under 120ms at scale.
- Achieved seamless deployment with Docker, Kubernetes, Istio service mesh, and AWS services. Implemented multi-region active-passive deployment with automated failover achieving 99.95%+ availability. Rigorous load testing using Java 21 virtual threads optimized high throughput and low latency.

◆ **Tech:** Java 21 Virtual Threads, Spring Boot, DocumentDB, OpenSearch, Redis, Kafka, WebSocket, Istio, AWS, OpenTelemetry, Prometheus, Grafana, Kubernetes

### Senior Software Engineer (Full Stack)

**SAP Concur** • *Mar 2017 – Feb 2022,* Bangalore, India

**System Integration Platform – Backend & Orchestration**

- Architected and delivered highly available Spring Boot microservices on Kubernetes for SAP Concur's enterprise integration platform — serving thousands of global enterprise customers. Built the Orchestration Framework transforming declarative customer configurations into distributed data-processing jobs used daily by 100+ integration consultants.

**Web Studio – Integration Flow Designer**

- Led the frontend team building Web Studio (React) — a visual flow designer with real-time WebSocket-streamed execution debugging for integration consultants. Delivered automated zero-touch monthly patching pipeline replacing a 3-day manual process across multiple customers deployments.

◆ **Tech:** Java/J2EE, Spring Boot, React, WebSocket, MySQL, Kubernetes, Helm, Jenkins, AWS (EC2, S3, RDS, SQS)

### Software Engineer (Full Stack)

**ComicFlix INC.** • *Jul 2014 – Mar 2017,* Bangalore, India

**Comify – Video to Comic Conversion Platform**

- Built Comify end-to-end — dialogue detection, scene boundary detection, speaker identification, and OpenCV cartoon rendering pipeline with near-zero manual intervention. Led a team of 4 across backend and frontend.

◆ **Tech:** Java, Spring Boot, Python, OpenCV, MySQL, GIMP, FFMPEG, JavaScript, jQuery

stations using LLM-driven route optimisation.

**Koni – Nearby Ask**
Hyperlocal platform for tier 2–3 cities: post a need, nearby shops/providers respond instantly.

## FIND ME ONLINE


🔗 **LinkedIn**
linkedin.com/in/upkar-kumar
**GitHub**
github.com/redhat125
🌐 **Portfolio**
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## EDUCATION

**B.E. – Computer Science Engineering**
**Dayananda Sagar College of Engineering**
Bangalore, Karnataka | 2010 – 2014